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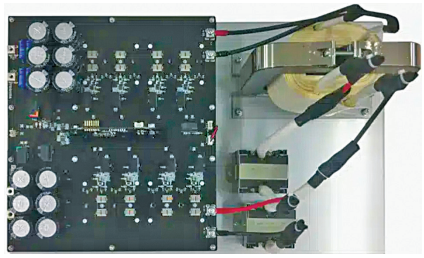
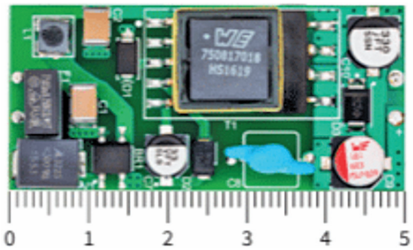
Isolated 40W Power Supply	5kW Isolated DC-DC Power Supply	Off-line Isolated Power Supply
		
The reference design transforms 24V into various voltage rails, features integrated control circuitry, and utilises an active-clamp transformer reset topology, ensuring compactness, efficiency, and reduced electromagnetic interference.	The reference design showcases a dual active bridge method, enabling efficient power transfer and voltage conversion with enhanced efficiency and advanced control features for optimising EMI noise and system performance.	The reference design utilises a current-mode flyback controller with built-in voltage capabilities, ensuring low power consumption, high efficiency, and robust protection features.
Maxim's isolated power supply reference design, known as MAXREFDES116#, converts 24V into various voltage rails using the MAX17599 PWM controller. This controller integrates control circuitry for isolated forward-converter industrial power supplies and operates across a 17V to 36V input voltage range, delivering up to 8A at 5V output. The design uses an active-clamp transformer reset topology for forward converters, offering benefits like reduced voltage stress on switching elements, the ability to use smaller transformers due to a higher permissible flux swing, and improved efficiency by eliminating the need for snubber circuits. These features result in a more compact and cost-effective power supply. Operating at a frequency of 350kHz, the design includes a feature to minimise electromagnetic interference for spread-spectrum operation.	Toshiba's 5kW isolated bidirectional DC-DC converter reference design efficiently delivers up to 5kW of power, capable of transferring voltage from the high to the low side and vice versa using the dual active bridge (DAB) method. This method utilises a full-bridge configuration on both sides, offering greater power handling than the half-bridge method and enabling soft switching through phase shifting, which enhances efficiency. The gate driver, incorporating the TLP5214A smart gate driver coupler with overcurrent protection and a UVLO function, drives the 1200V SiC MOSFET with a 4A sink-source current capability. The gate drive circuit, crucial for balancing power supply efficiency and EMI noise, allows adjustable MOSFET switching speeds, and employs strategies like increasing gate series resistance to reduce noise, which also impacts the MOSFET's switching speeds and overall system efficiency. Adjustments to gate series resistance are carefully managed to maintain a balance between EMI noise, efficiency, and heat dissipation.	The reference design features the Infineon CoolSET ICE3RBR4765JG as a current-mode flyback controller with a built-in 650V CoolMOS, operating in discontinuous conduction mode (DCM) at a switching frequency of 65kHz to produce a 5V, 600mA output. It incorporates active burst mode operation for extremely low standby power consumption, less than 13mW, across an input voltage range of 180–265V AC, and achieves low EMI through built-in frequency jitter and soft start operation. The design includes protection modes for Vcc over/undervoltage, IC overtemperature, overload, and open loop. This efficient and reliable solution offers low power consumption, high efficiency, robust protection features, and compliance with EMI standards, making it ideal for powering a wide range of IoT devices and modern electronic designs.
The reference design is suitable for PLCs, industrial process control, industrial sensors, and telecom/datacom power supplies.	The reference design is suitable for EV charging systems and photovoltaic power generation inverters.	The reference design is suitable for various IoT applications.
Maxim	Toshiba	Infineon Technologies
https://www.electronicsforu.com/electronics-projects/isolated-40w-power-supply-reference-design	https://www.electronicsforu.com/electronics-projects/5-kw-isolated-dc-dc-power-supply-reference-design	https://www.electronicsforu.com/electronics-projects/reference-design-for-off-line-isolated-power-supply